

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	6 July 2026
Team ID	
Project Name	BookStore
Maximum Marks	4 Marks

Technical Architecture:

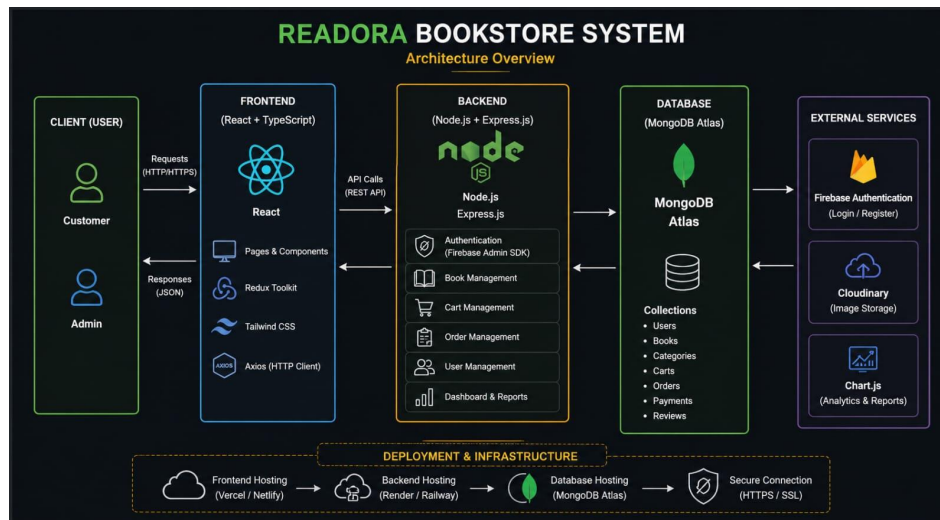
Readora – MERN Bookstore is a full-stack web application developed using the MERN Stack (MongoDB, Express.js, React, and Node.js). The application provides a secure and user-friendly platform where users can browse books, search by category, manage shopping carts, place orders, and securely authenticate using Firebase Authentication. Administrators can manage books, monitor sales statistics, and analyze revenue through an interactive dashboard.

The frontend is developed using React with TypeScript and Tailwind CSS for building a responsive user interface. The backend is built with Node.js and Express.js, exposing RESTful APIs that communicate with MongoDB Atlas for data storage. Redux Toolkit is used for state management, while Chart.js is used to visualize dashboard analytics.

The architecture follows the MVC (Model–View–Controller) design pattern to ensure modularity, maintainability, and scalability.

Example: Readora – MERN BookStore

Reference: <https://react.dev/>
<https://www.mongodb.com/docs/>



Guidelines:

Illustrate the complete architecture of the Readora Bookstore system, including the React Frontend, Express.js Backend, MongoDB Atlas Database, Firebase Authentication, and REST API communication.

Clearly represent how users interact with the frontend, how requests are processed through the backend, how data is stored in MongoDB Atlas, and how authentication and administrative operations are securely managed.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive web application for Customers & Admin	React.js, TypeScript, Tailwind CSS
2.	Authentication	Secure Login & Registration	Firebase Authentication
3.	Application Logic	Book Browsing, Search, Cart & Orders	Node.js, Express.js
4.	Admin Module	Book Management & Dashboard	React.js, Express.js
5.	Database	Stores Books, Users & Orders	MongoDB
6.	Cloud Database	Cloud Hosted Database	MongoDB Atlas
7.	State Management	Global Application State	Redux Toolkit
8.	Data Visualization	Dashboard Analytics	Chart.js

9.	REST API	Communication between Frontend & Backend	Express.js REST API
10.	Infrastructure	Backend Server Hosting	Node.js Server / Localhost
11.			

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Secure user authentication and protected admin routes	Firebase Authentication
3.	MVC Architecture	Modular project structure following MVC pattern	Express.js MVC
4.	Availability	Application accessible 24×7 with cloud database	MongoDB Atlas
5.	Performance	Fast API response and optimized database operations	Express.js, MongoDB
1.	Responsive Design	Supports desktop, tablet and mobile devices	Tailwind CSS
2.	Scalability	Easily supports increasing users, books and orders	MERN Stack
3.	Maintainability	Clean, modular and reusable codebase	TypeScript, MVC

References:

<https://react.dev/>

<https://nodejs.org/>

<https://expressjs.com/>

<https://www.mongodb.com/docs/>

<https://firebase.google.com/docs/auth>

<https://www.chartjs.org/>

<https://tailwindcss.com/>

<https://redux-toolkit.js.org/>